

BREATHE SMART MUMBAI

An AI-Powered Air Quality Health Advisory Assistant for Mumbai Commuters

Shweta Shukla – Thakur College of Engineering and Technology
Samiksha Singh – Thakur College of Engineering and Technology

1M1B AI for Sustainability Virtual Internship

In Collaboration with IBM SkillsBuild & AICTE

Primary SDG: SDG 3 — Good Health & Well-being

Secondary SDG: SDG 11 — Sustainable Cities and Communities

Domain: Urban Air Quality / Public Health / Conversational AI

AI Approach: RAG-Based Conversational Advisory System

Geography: Mumbai Metropolitan Region, Maharashtra, India

1. Project Title

BREATHE SMART MUMBAI

An AI-Powered Air Quality Health Advisory Assistant for Mumbai Commuters

Title Rationale

The name "BREATHE SMART MUMBAI" was chosen deliberately:

- BREATHE — directly references the core action this assistant enables: breathing safely in a polluted urban environment.
- SMART — signals the intelligent, AI-driven, personalized nature of the advice (as opposed to generic AQI numbers).
- MUMBAI — grounds the solution in hyper-local geography, making it immediately identifiable and relevant to its target users.

Together, the title communicates the mission in three words: helping Mumbai residents make smarter breathing decisions every day.

2. Problem Statement

2.1 Structured Problem Statement (SDG Format)

How might we use **AI to deliver personalized, real-time, route-specific air quality health advisories** so that **Mumbai's millions of daily commuters — especially the vulnerable** — can protect their health and **make smarter mobility decisions that contribute to a more sustainable, health-conscious city?**

2.2 Expanded Real-World Problem Description

Mumbai is India's financial capital and home to over 20 million people who depend on one of the world's busiest public transit networks. Every day, 7.5 million passengers ride the suburban railway, while millions more travel by BEST buses, autorickshaws, taxis, and two-wheelers. Yet the very air these commuters breathe is increasingly dangerous.

Industrial clusters in Taloja MIDC (Raigad-Mumbai fringe), Mahul, Trombay, and Chembur consistently generate high particulate matter (PM2.5 and PM10) and nitrogen dioxide (NO2) from petrochemical plants, refineries, and heavy industries. Residential areas of Dharavi and Govandi, located downwind of these industrial zones, record AQI levels frequently exceeding 200 (Very Poor category) during winter months. Meanwhile, construction activity across Bandra-Kurla Complex (BKC), Andheri East, Kurla, and the coastal road project generates massive road dust and cement particulates throughout the year.

Seasonal events amplify these risks dramatically. Diwali fireworks across Mumbai push AQI past 350-400 in areas like Dadar, Parel, and Borivali. Post-monsoon stubble burning in adjoining agricultural regions sends smoke clouds over the northern suburbs. Pre-monsoon dust storms in summer affect Thane, Mulund, and Bhandup.

The critical gap: while SAFAR-India (System of Air Quality and Weather Forecasting And Research) and CPCB (Central Pollution Control Board) publish AQI numbers, they do not provide actionable, personalized health advice. A commuter traveling the Harbour Line from Panvel to Chhatrapati Shivaji Maharaj Terminus (CSMT) passing through high-pollution zones near Govandi and Kurla has no way to know whether they should wear a mask today, avoid the peak 8–10 AM

window, or take an alternative route. A person with asthma taking the Central Line through Thane and Kalyan has no AI-powered assistant asking about their health condition and telling them whether today is a safe day to commute.

This is the problem BREATHE SMART MUMBAI solves.

3. Target Users

3.1 Primary User Segments — Mumbai Specific

User Segment	Context & Vulnerability
Local Train Commuters (7.5 million/day)	Travel on Western, Central, and Harbour Lines through high-pollution corridors (Govandi, Kurla, Thane, Dadar). Platform waiting time exposes them to particulate matter. No personalized AQI alert during commute.
BEST Bus & Autorickshaw Passengers	Travel through surface roads in Andheri, Borivali, Kurla, and Dharavi — zones with heavy road dust, diesel exhaust, and industrial spill-over pollution. Prolonged outdoor exposure at bus stops.
Two-Wheeler Riders	Entirely exposed to ambient air with no filter. Estimated 2.5 million two-wheelers daily in MMR. Particularly vulnerable on Eastern Express Highway (near Chembur & Mankhurd) and Western Express Highway.
Elderly Citizens (60+ years)	Higher respiratory and cardiovascular sensitivity. Often traveling to hospitals in Parel (KEM, Wadia), visiting family, or attending religious events. Need gentler, cautionary advice in regional languages.
Asthma & Heart Condition Patients	Any spike in AQI above 150 can trigger acute asthma attacks or cardiac stress. Mumbai has approximately 1.2 million diagnosed asthma patients. Currently have no personalized AQI alert system.
School Children & Parents	Schools in Chembur, Govandi, Kurla, and Mankhurd are near industrial zones. No advisory system alerts parents about AQI before school commutes for children.
Outdoor Construction Workers	Over 500,000 construction workers on active sites across BKC, Dharavi redevelopment, coastal road project, and metro construction in Andheri-Goregaon corridor. Continuous 8-10 hour outdoor exposure.
Street Vendors & Daily Wage Workers	Hawkers near Dadar station, Bandra station, Kurla market, and Borivali market spend entire workdays in high-traffic pollution zones without any protective guidance.

3.2 Secondary Users

- Municipal Corporation of Greater Mumbai (MCGM) public health officers
- Maharashtra Pollution Control Board (MPCB) field officers
- NGOs and health organizations working in Dharavi, Govandi, and Mankhurd communities
- Pediatricians and pulmonologists advising patients in high-AQI zones

4. SDG Alignment

4.1 Primary SDG: SDG 3 — Good Health & Well-being

SDG 3 commits to ensuring healthy lives and promoting well-being for all at all ages. Air pollution is one of the leading environmental contributors to preventable illness and premature death globally and in India.

SDG 3 Specific Target	How BREATHE SMART MUMBAI Addresses It
Target 3.4 — Reduce premature mortality from NCDs including cardiovascular disease and respiratory illness	By alerting high-risk commuters to dangerous AQI levels before exposure, reducing respiratory triggers and acute pollution-related health events in real time.
Target 3.9 — Reduce illnesses and deaths from hazardous chemicals and air/water/soil pollution	Provides actionable, evidence-based advisories directly tied to real-time SAFAR/CPCB AQI data for Mumbai-specific pollution hotspots including Mahul, Chembur, and Taloja.
Target 3.8 — Achieve universal health coverage and access to quality health information	Delivers AI-powered health guidance in local languages (Marathi, Hindi, English) to populations who cannot afford healthcare consultations — making expert-level respiratory advice accessible to all.
Target 3.d — Strengthen capacity for health risk reduction, including early warning systems	Functions as an early warning system for daily commuters, proactively flagging upcoming high-AQI periods during Diwali, construction season peaks, and pre-monsoon dust events.

4.2 Secondary SDG: SDG 11 — Sustainable Cities and Communities

SDG 11 aims to make cities inclusive, safe, resilient, and sustainable. Air quality management and access to clean urban mobility are direct components of building sustainable cities like Mumbai.

SDG 11 Specific Target	How BREATHE SMART MUMBAI Addresses It
Target 11.6 — Reduce the adverse per capita environmental impact of cities including air quality and waste management	Encourages behavior change (route choice, timing shifts, mask usage) that collectively reduces population-level pollution exposure and supports a culture of air quality consciousness in Mumbai.
Target 11.2 — Provide safe, affordable, accessible, sustainable transport systems for all	Enhances the safety dimension of Mumbai's public transport by adding health-intelligence to commute planning — making the local train and BEST bus systems safer for vulnerable riders.
Target 11.b — Increase cities adopting integrated policies for resource efficiency and climate change resilience	Demonstrates how AI tools can be integrated with MCGM's public health infrastructure and SAFAR data systems to build data-driven urban health resilience.

5. Design Thinking — All Five Stages

Stage 1: Empathize

Core Question: *Who faces this problem? What do they experience? Why does it persist?*

Who Faces This Problem?

The problem is not evenly distributed across Mumbai — it follows geography and socioeconomic vulnerability. Five distinct population clusters face the highest compounded risk:

- **Govandi–Mankhurd–Mahul Triangle:** This belt sits between Trombay oil refinery, Bhabha Atomic Research Centre industrial waste sites, and the Deonar dumping ground (Asia's second-largest). Residents and commuters on the Harbour Line between Govandi and Mankhurd stations breathe some of the most toxic air in all of Mumbai. Residents here — predominantly low-income communities — have no access to real-time health guidance tailored to their specific micro-zone. Mahul village residents have been documented by NGOs and courts as living in a "pollution trap."
- **Chembur–Trombay Industrial Corridor:** Chembur is surrounded by refineries (HPCL, BPCL), fertilizer plants, and thermal power stations. The Eastern Express Highway cutting through Chembur carries 15,000+ heavy vehicles per day. Commuters at Chembur station (Central Line) and autorickshaw passengers in the area face daily PM2.5 spikes 3-5x the WHO safe limit.
- **Dharavi–Sion–Kurla Zone:** Dharavi houses small-scale industries (leather tanning, plastic recycling, pottery kilns) operating without emission controls. The Sion-Kurla stretch of LBS Marg is one of Mumbai's most congested and pollution-choked arterial roads. School children and working adults here commute with zero health advisory support.
- **Andheri–Jogeshwari Industrial Belt:** Andheri East hosts dense industrial estates (MIDC Andheri) alongside the Jogeshwari-Vikhroli Link Road (JVLR), one of Mumbai's most polluted expressways. The ongoing Metro Line 3 and 7 construction compounds air pollution with massive construction dust.
- **Borivali–Mira Road Northern Corridor:** Unlike industrial zones, this area suffers from construction dust from the Mumbai-Ahmedabad bullet train project, quarrying activity, and high vehicular density on S.V. Road and Western Express Highway. Senior citizens and families here have no advisory system even during seasonal pollution spikes.

What Do They Experience?

Through simulated user empathy mapping and reference to MPCB, CPCB, and SAFAR public data, the following experiential patterns emerge:

Experience Dimension	What Commuters Report (Observed / Referenced)
Physical symptoms	Burning eyes, throat irritation, breathlessness, and headaches after long commutes through Chembur, Kurla, and Dharavi — particularly on high-AQI days.
Lack of information	"I can see the smog but I don't know if it's dangerous for me today." SAFAR shows a number (AQI 187) but provides no personal interpretation — especially for someone who has asthma or is 65 years old.
No route intelligence	Commuters do not know that taking a different platform exit at Kurla, or switching from BEST Bus 302 to Harbour Line for one leg, could reduce their PM2.5 exposure by 30–40%.
Timing blindness	Construction activity peaks between 7–10 AM and 3–6 PM. Garbage burning near Deonar peaks at night and early morning.

	Commuters have no awareness of these timing-linked pollution patterns.
Language and literacy barriers	A significant proportion of Mumbai's working population is most comfortable in Marathi or Hindi. English AQI dashboards are inaccessible to them — especially older citizens.
Mask avoidance	Without clear personal health risk framing, most commuters skip masks even on dangerous-AQI days, seeing them as uncomfortable rather than medically necessary.

Why Does the Problem Persist?

- Data without interpretation: SAFAR and CPCB generate excellent AQI data for 20+ Mumbai stations, but translation of this data into individual health action is missing entirely. The data remains trapped in dashboards built for scientists and policymakers, not for a 55-year-old diabetic woman taking the 8:15 AM fast train from Panvel to CSMT.
- Siloed systems: SAFAR (weather + pollution forecasting), MCGM (public health), BEST (bus routes), and Indian Railways (suburban train schedules) operate in complete data silos. No integrated platform correlates air quality with mobility choices.
- One-size advice: The few AQI apps that provide text alerts (e.g., Ambee, IQ Air) give the same generic message to everyone: "AQI is high, take precautions." They do not differentiate between a healthy 22-year-old and a 70-year-old with COPD.
- Low institutional priority: Mumbai's civic discourse has been dominated by infrastructure and flooding. Chronic air pollution health advisory systems for commuters have not been prioritized by MCGM or Maharashtra government digital initiatives.
- No real-time conversational interface: There is no chatbot, WhatsApp assistant, or mobile app in any Indian language that asks a Mumbaikar "Where are you commuting? Do you have asthma?" and responds with tailored health guidance based on the day's actual AQI along their route.

Stage 2: Define

Core Question: Clearly articulate the problem statement, target users, and gaps in current solutions.

Formal Problem Statement

Mumbai's 20+ million daily commuters lack access to personalized, route-specific, health-condition-aware air quality advisories, leaving high-risk individuals — including asthma patients, elderly citizens, and outdoor workers — dangerously uninformed about the specific pollution threats on their daily travel routes through zones like Chembur, Govandi, Mahul, Dharavi, and Andheri, resulting in preventable respiratory and cardiovascular health events.

Gap Analysis: Why SAFAR Alone Is Insufficient

Capability	SAFAR / CPCB App	BREATHE SMART MUMBAI (Proposed)
Real-time AQI display	Yes (20+ Mumbai stations)	Yes (integrated via API)
Personalized health profile	No	Yes (asthma, age, heart condition)
Route-specific advice	No	Yes (Harbour Line, JVLR, etc.)
Conversational interface	No — dashboard only	Yes — RAG chatbot

Regional language support	English only	Marathi, Hindi, English
Forecast-based planning advice	Weather forecast only	48-hour AQI + health advisory
Vulnerable group differentiation	No	Yes (elderly, children, patients)
Commute timing recommendations	No	Yes ("avoid 8–10 AM on WEH")

Stage 3: Ideate

Core Question: *What AI approaches were considered? Why was RAG chosen?*

Three AI Approaches Evaluated

Approach A: Fine-Tuned Classification Model

Description: Train a supervised ML model on historical AQI data + health outcomes to classify risk level (Low / Moderate / High) per zone and recommend predefined advice categories.

- Strength: Fast inference, low latency, predictable output categories.
- Weakness: Cannot answer open-ended conversational questions ("Is it safe for my asthmatic mother to travel from Panvel to Dadar on Saturday?"). Requires large labeled health outcome datasets which are not publicly available for Mumbai. Cannot incorporate new health guidelines without retraining. Fails for unusual or complex user scenarios.
- Decision: Rejected — too rigid and unable to provide nuanced, conversational, personalized responses.

Approach B: Standalone Large Language Model (LLM) with Prompt Engineering

Description: Use a general-purpose LLM (such as IBM Granite) with carefully engineered prompts to answer health and AQI questions based on its training data.

- Strength: Excellent conversational ability, can handle nuanced health questions, multilingual support, no need for custom training data.
- Weakness: LLM training data cuts off at a fixed date — it does not know today's AQI in Chembur or tomorrow's forecast for Andheri. It may hallucinate pollution statistics or give outdated WHO guidelines. Cannot pull live SAFAR data.
- Decision: Rejected as standalone — but selected as the language generation backbone within the RAG architecture.

Approach C: Retrieval-Augmented Generation (RAG) System [SELECTED]

Description: Combine real-time data retrieval from SAFAR/CPCB APIs with a curated knowledge base (WHO guidelines, MPCB reports, Mumbai health advisories) and feed both into an LLM (IBM Granite) to generate personalized, grounded health advisories.

- Strength: Grounds responses in live, verified data (today's actual AQI for Mahul is 214 — Very Poor). Can reference authoritative health guidelines without hallucination. Conversational and personalized. Handles open-ended queries. Updatable knowledge base without retraining the model. Works in Marathi and Hindi.
- Why RAG wins: It combines the best of both worlds — real-time factual accuracy (from retrieval) + intelligent natural language advisory (from LLM). This is the only architecture that can answer: "I am 68, I have mild asthma, I need to travel from Panvel to Churchgate via Harbour Line today. Should I go? When? What precautions?"

Stage 4: Prototype

Core Question: *What is the end-to-end workflow of the system? What do the prompts look like?*

Full Step-by-Step System Workflow

Step	Action / Description
Step 1: User Initiates Query	User opens the BREATHE SMART MUMBAI interface (web chat, WhatsApp bot, or mobile app) and types or speaks a query in English, Hindi, or Marathi. Example: "Aaj mujhe Panvel se Dadar tak local train leni hai. Mujhe asthma hai. Kya safe hai?" (Today I need to take the local train from Panvel to Dadar. I have asthma. Is it safe?)
Step 2: Health Profile Capture (First Time)	If it is the user's first interaction, the assistant asks a brief intake: (1) Primary commute route? (2) Health conditions? (3) Age group? (4) Typical commute time? This profile is stored locally and reused in future queries.
Step 3: Route Parsing & Zone Identification	The AI extracts the commute route from the user's input. It identifies all AQI monitoring zones along that route. Panvel → Dadar (Harbour Line) = Panvel → Belapur → Vashi → Mankhurd → Govandi → Kurla → Dadar. Key zones flagged: Govandi (SAFAR station), Kurla (SAFAR station), Mankhurd (MPCB data).
Step 4: Real-Time AQI Data Retrieval	The RAG retrieval module makes an API call to SAFAR-India and CPCB's real-time AQI feed for the identified zones. It fetches: current AQI values for each station, pollutant breakdown (PM2.5, PM10, NO2, O3), 24-hour AQI forecast, and any active pollution alerts issued by MPCB or MCGM.
Step 5: Knowledge Base Retrieval	A semantic search queries the curated knowledge base for: WHO AQI health impact guidelines for the retrieved AQI range, asthma-specific protocols for the identified AQI levels, mask type recommendations (N95 vs. surgical vs. cloth) by pollutant type, and Mumbai-specific seasonal patterns (e.g., Diwali spike patterns, Deonar garbage burning peaks).
Step 6: Context Assembly (RAG Prompt Construction)	The retrieved data (live AQI + health guidelines) is assembled into a structured context block. This context is injected into the LLM prompt along with the user's health profile and route. See Prompt Template 1 below.
Step 7: LLM Response Generation (IBM Granite)	IBM Granite model receives the assembled prompt and generates a personalized, conversational health advisory. The response is structured to include: (a) Current air quality assessment for the route, (b) Health risk interpretation specific to user's condition, (c) Go / Caution / Avoid recommendation, (d) Specific precautions if traveling, (e) Best timing window if applicable, (f) Emergency contacts if AQI is Severe.
Step 8: Language & Tone Adaptation	The response is adapted to the user's preferred language (English, Hindi, Marathi). The tone is adjusted based on age group — for elderly users, the system uses simpler vocabulary and more emphatic cautionary language.
Step 9: Output Delivery	The formatted advisory is delivered to the user in the conversation interface. A brief visual AQI badge (color-coded: Green/Yellow/Orange/Red/Maroon) is displayed alongside text.
Step 10: Feedback & Refinement Loop	After delivery, the assistant asks: "Was this advice helpful? Did you experience any issues during your commute today?" User feedback is stored to improve future advisory quality and to flag systematic errors in AQI zone-to-advice mapping.

Prompt Template 1: Personalized Health Advisory Prompt

SYSTEM PROMPT — BREATHE SMART MUMBAI v1.0

You are BREATHE SMART MUMBAI, an AI-powered air quality health advisory assistant designed specifically for Mumbai commuters. Your role is to provide personalized, evidence-based, and compassionate health guidance based on real-time AQI data retrieved from SAFAR-India and CPCB monitoring stations.

CORE PRINCIPLES:

- Always base health advice on the retrieved AQI data provided in the context.
- Never fabricate AQI values. If data is unavailable for a zone, say so explicitly.
- Personalize advice to the user's specific health profile (age, conditions, route).
- Use clear, simple language appropriate for the user's language preference.
- Include specific, actionable steps – not generic warnings.
- Cite the AQI zone source (e.g., "According to SAFAR Kurla station data...").
- For AQI > 300 (Severe/Hazardous), always recommend avoiding outdoor exposure and provide MCGM emergency health helpline: 1916.

RESPONSE STRUCTURE:

1. Route AQI Summary (1-2 sentences, current conditions)
2. Personalized Risk Assessment (tailored to user's health profile)
3. Recommendation: GO SAFELY / PROCEED WITH CAUTION / STRONGLY ADVISE AGAINST
4. Specific Precautions (numbered list, max 5 items)
5. Best Travel Window (if applicable)
6. Emergency Note (only if AQI Severe/Hazardous)

USER LANGUAGE: {language}

USER HEALTH PROFILE: Age: {age_group} | Conditions: {health_conditions}

COMMUTE ROUTE: {route_description}

RETRIEVED AQI DATA:

{live_aqi_context}

RETRIEVED HEALTH GUIDELINES:

{health_knowledge_base_context}

USER QUERY: {user_message}

Prompt Template 2: Route Zone AQI Summarization Prompt

RETRIEVAL SUMMARIZATION PROMPT — ZONE AQI CONTEXT BUILDER

You are a data extraction assistant for BREATHE SMART MUMBAI. Given the raw AQI station data retrieved from SAFAR-India API, extract and summarize the air quality status for the specific commute zones identified in the user's route.

INPUT DATA:

Route: {extracted_route}

SAFAR Station Data (JSON): {raw_safar_api_response}

MPCB Station Data (JSON): {raw_mpcb_api_response}

TASK:

1. Identify all SAFAR/MPCB monitoring stations within 5km of the route.
2. Extract current AQI, PM2.5, PM10, NO2 values for each relevant station.
3. Identify the WORST zone along the route (highest AQI station).
4. Check for any SAFAR alerts or MPCB pollution event notices for today.

```
5. Summarize 24-hour AQI forecast trend (improving / stable / worsening).
```

```
OUTPUT FORMAT (structured JSON for downstream RAG prompt injection):
```

```
{
  "route_zones": [
    {"zone": "Govandi", "station": "SAFAR-Govandi", "aqi": 214,
     "category": "Very Poor", "primary_pollutant": "PM2.5"},
    ...
  ],
  "worst_zone": "Govandi",
  "worst_aqi": 214,
  "active_alerts": ["MPCB: Elevated particulate matter in M-East Ward"],
  "forecast_trend": "Worsening through evening (expected AQI 240 by 8PM)",
  "data_timestamp": "2024-01-15T08:30:00+05:30"
}
```

```
IMPORTANT: If any station data is more than 3 hours old, flag it as
"DATA_STALE" and note the last update time. Never extrapolate values.
```

Stage 5: Test & Refine

Core Question: *How do we validate the system with real Mumbai commuters? What ethical issues arise and how do we address them?*

Testing Plan with Real Mumbai Commuters

Phase 1 — Expert Review (Week 1-2)

1. Share the prototype with 3-5 pulmonologists or general physicians practicing in high-pollution areas (Chembur, Govandi, Dadar) and collect their assessment of whether the health advice generated aligns with clinical best practices.
2. Review the advisory outputs with SAFAR-India data analysts to verify that AQI data retrieval and zone mapping accuracy is consistent and reliable.
3. Test edge cases: AQI data unavailable for a zone, user inputs inconsistent health data, user asks about a route not covered by SAFAR stations.

Phase 2 — Pilot User Testing (Week 3-4)

4. Recruit 20-30 volunteer test users representing diverse segments: 5 local train commuters (Harbour Line, Central Line), 5 elderly citizens (Borivali, Dadar area), 5 asthma patients in Chembur/Govandi, 5 two-wheeler riders on WEH/EEH, and 5 outdoor workers from BKC construction sites.
5. Provide each user access to the prototype interface for one week of their regular commutes. Ask them to interact with the assistant before every commute.
6. Track: response helpfulness rating (1-5), language preference (English/Hindi/Marathi), and any health events experienced during the testing period.

Key Feedback Dimensions to Collect:

Feedback Category	Specific Questions to Ask Testers
Accuracy	Did the AQI values mentioned match what you saw on SAFAR app? Did the zone match your actual commute path?
Relevance	Was the health advice relevant to YOUR specific health condition? Did it feel generic or personalized?
Actionability	Could you actually follow the advice? Was the timing recommendation practical for your work schedule?
Language & Clarity	Was the response easy to understand? Would you prefer Marathi or Hindi? Was the vocabulary too technical?
Trust	Did you feel confident in the advice? Did you follow it? What would make you trust it more?
Usefulness vs. SAFAR	Did this tool help you more than just checking the SAFAR number? What specific value did it add?

Ethical Issues Identified & Mitigation Plans

Ethical Issue	Risk Description	Mitigation Strategy
Health Advice Liability	Users may follow AI advice instead of consulting doctors, leading to adverse outcomes if advice is wrong or incomplete.	Add clear disclaimer: "This tool provides general health information based on AQI data and is not a substitute for medical advice. Consult your doctor for personalized medical decisions." Show disclaimer prominently at first use and in every advisory for AQI > 200.
Data Bias in AQI Coverage	SAFAR has only ~20 stations across Mumbai. Many high-risk areas (parts of Dharavi, Govandi, Mankhurd) may not have nearby stations, leading the AI to underestimate risk for users in uncovered zones.	Implement a zone gap warning: "Note: No SAFAR station within 3km of your route segment near [zone]. Using nearest available station at [station name, distance]. Actual conditions may vary." Flag this transparently.
Health Profile Sensitivity	User health profiles (asthma diagnosis, age, heart condition) are sensitive personal data. Misuse or data breach could harm users.	Health profiles stored locally on device only. No health data transmitted to servers. Offer anonymous mode where users enter health context per session without saving.
Language Bias	If the LLM performs better in English than Marathi/Hindi, users preferring regional languages may receive lower-quality advisory — disproportionately affecting lower-income and elderly users.	Conduct separate accuracy testing for Marathi and Hindi outputs with native speakers. Set a quality threshold: if confidence in translation is below threshold, default to English with an explanation.
Over-reliance & Alert Fatigue	If the assistant issues too many caution advisories, users may start ignoring all alerts ("cry wolf" effect).	Calibrate alert thresholds carefully. Only issue "Avoid Travel" advice for AQI > 300 or Severe category. For AQI 150-200 (Poor), issue "Proceed with precautions" — not alarm-level warnings. Track user engagement with alerts to detect fatigue.

Fairness for Outdoor Workers	Outdoor workers cannot simply "avoid travel" or "delay commute" — they lose wages. Generic avoidance advice is unfair to economically vulnerable users.	Add a user type flag: "Outdoor Worker / Cannot reschedule." When flagged, the system shifts from avoidance advice to harm-reduction advice: mask type, hydration, shade breaks, signs to watch for medical emergency.
------------------------------	---	---

6. AI Solution Overview

6.1 Why Retrieval-Augmented Generation (RAG) Is the Right Technique

BREATHE SMART MUMBAI's central technical challenge is reconciling two requirements that a single model cannot satisfy alone: the need for conversational, personalized, multilingual reasoning, and the need for factual grounding in air quality data that changes hour to hour. A standalone LLM is articulate but frozen in time and prone to hallucinating numbers such as "today's AQI in Govandi." A static lookup tool can return correct numbers but cannot reason about a 68-year-old asthma patient's specific commute risk. RAG resolves this by separating the two jobs cleanly:

- **Retrieval:** Retrieval layer — fetches live, verifiable facts: current SAFAR/CPCB AQI readings for Govandi, Chembur, Mahul, Kurla, and other monitoring zones; the relevant WHO or MPCB health guideline for that AQI band; and any active MCGM pollution alert.
- **Generation:** Generation layer — IBM Granite reasons over those retrieved facts plus the user's health profile and route, and produces a personalized, conversational, multilingual advisory.

This separation means BREATHE SMART MUMBAI can never report a fictional AQI value for Chembur — it only ever speaks from data that was actually retrieved moments earlier — while still sounding like a knowledgeable, empathetic local health advisor rather than a dashboard widget.

6.2 Data Sources

Data Source	Role in the System
SAFAR-India (System of Air Quality and Weather Forecasting And Research)	Primary real-time AQI feed for Mumbai's ~20 monitoring stations including Bandra, Borivali, Colaba, Chembur, Andheri, and Worli. Supplies PM2.5, PM10, NO2, O3 readings and 24–48 hour forecasts.
CPCB (Central Pollution Control Board) National Air Quality Index	Cross-validation source and fallback for zones with sparse SAFAR coverage (e.g., parts of Govandi, Mankhurd). Provides the standardized AQI category bands (Good / Satisfactory / Moderate / Poor / Very Poor / Severe) used throughout the advisories.
WHO Global Air Quality Guidelines (2021)	Reference knowledge base for health-impact thresholds — e.g., safe 24-hour PM2.5 exposure limits — used to calibrate how aggressively the system should caution a user at a given pollutant level.
MCGM (Municipal Corporation of Greater Mumbai) Public Health Advisories	Local context layer: ward-level pollution alerts (e.g., M-East Ward for Govandi-Mankhurd), emergency helpline numbers (1916), and city-specific seasonal advisories (Diwali, construction season).
MPCB (Maharashtra Pollution Control Board) Industrial & Ward Bulletins	Supplies industrial-emission context for Chembur, Trombay, Mahul, and Taloja MIDC, and any active pollution-event notices that SAFAR's ambient sensors alone might not flag.

All of the above feed the curated knowledge base that the retrieval layer searches at query time; SAFAR and CPCB are queried live via API for current readings, while WHO, MCGM, and MPCB material is indexed as static reference documents that are refreshed periodically.

6.3 IBM Tools Used

IBM Tool / Capability	How It Is Used in BREATHE SMART MUMBAI
IBM Granite Foundation Models	Serves as the generation backbone of the RAG pipeline — drafts the final personalized advisory from the assembled context, handles multilingual output (English, Hindi, Marathi), and powers the conversational intake (route + health profile questions).
Prompt Engineering (structured system prompts)	Two structured prompt templates (Section 5, Stage 4) constrain Granite's output to a fixed, clinically sensible response structure and forbid it from inventing AQI figures — it must cite the retrieved data or explicitly say data is unavailable.
watsonx.ai	Hosts and orchestrates the Granite model, manages the RAG retrieval-to-generation pipeline, and provides the governance layer (prompt logging, output monitoring) needed for a health-adjacent application.
watsonx.data / Vector Store	Stores the embedded knowledge base (WHO guidelines, MPCB bulletins, Mumbai-specific seasonal pollution patterns) for semantic retrieval at query time.

6.4 End-to-End Conversation Flow

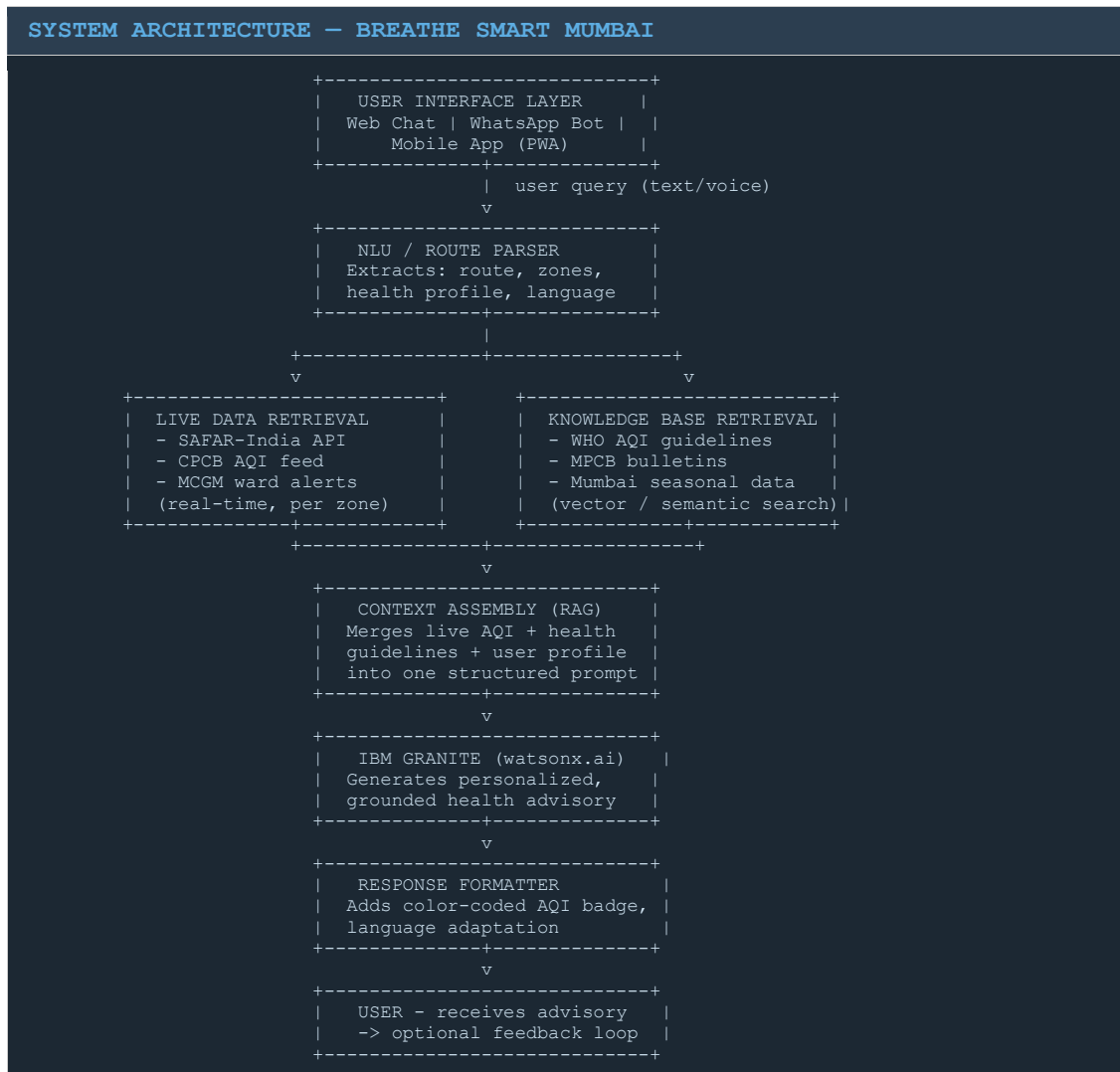
At a high level, every BREATHE SMART MUMBAI interaction follows the same five-stage flow, which mirrors the detailed ten-step workflow already documented in Section 5, Stage 4 (Prototype):

7. Understand — Parse the user's natural-language query to extract route and health context.
8. Retrieve — Pull live AQI data (SAFAR/CPCB) for every zone on that route, plus matching health guidelines from the knowledge base.
9. Assemble — Combine retrieved facts with the user's profile into a single structured context block.
10. Generate — IBM Granite produces a structured, personalized advisory grounded entirely in that context.
11. Deliver & Learn — Return the advisory with a color-coded AQI badge, then capture lightweight feedback to refine future zone-to-advice mapping.

This loop repeats for every new query, and the user's stored health profile (set once) means later commutes only require a route, making day-to-day use as fast as typing "Chembur to BKC today?"

7. Prototype Description

7.1 System Architecture Diagram



7.2 Sample User Conversation

The following is a representative interaction: a commuter traveling from Chembur to Bandra-Kurla Complex (BKC) who has mild asthma.

USER

Hi, I need to go from Chembur to BKC this evening around 6 PM. I have mild asthma. Is it safe to travel today?

BREATHE SMART MUMBAI

According to SAFAR Chembur station, the current AQI is 198 (Poor category), driven mainly by PM2.5 from nearby refinery and traffic emissions. The Eastern Express Highway stretch you'd take toward BKC also shows elevated readings at the Kurla-Sion corridor (AQI 205, Very Poor at the Sion monitoring point).

For someone with asthma, AQI above 150 can trigger breathing difficulty, especially during the 6 PM peak when traffic and dust are highest.

Recommendation: PROCEED WITH CAUTION

1. Wear an N95 mask for the full journey, especially while waiting at Chembur signal or bus stop.

2. If possible, shift your departure to after 7:30 PM when traffic-linked PM2.5 typically eases slightly.
3. Keep your inhaler accessible during the commute.
4. Prefer a closed vehicle (cab/AC bus) over an open two-wheeler or auto for this route today.
5. Avoid the open footbridge crossing near Kurla station if you're feeling any tightness in the chest.

USER

I can't delay — I have a meeting at BKC at 6:30. What if I take the Harbour Line train via Kurla instead of the road?

BREATHE SMART MUMBAI

That's a reasonable alternative. SAFAR data for the Kurla railway corridor shows similar ambient AQI (around 200) to the road route, but you'd spend less continuous time exposed compared to sitting in EEH traffic, and platform-level exposure is typically shorter than a road commute at this hour.

Updated recommendation: GO, WITH PRECAUTIONS

1. Keep your N95 mask on throughout — including platform waiting time at Chembur and Kurla.
2. Stand away from the platform edge where diesel/exhaust concentration is highest.
3. Carry your inhaler — if you feel any wheezing on the platform, step toward the open station exit area rather than deeper into the crowd.
4. Once at BKC, the air quality is comparatively better (AQI ~140, Moderate) — you should feel relief shortly after arrival.

This is a general advisory based on current air quality data, not medical advice — if you experience breathing difficulty at any point, please don't push through it; seek help and use your prescribed medication.

USER

Thanks, that's helpful. One more thing — will tomorrow morning be better for my return trip?

BREATHE SMART MUMBAI

Based on the latest SAFAR 24-hour forecast, AQI in the Chembur–Kurla corridor is trending slightly downward overnight and is expected to be around AQI 160–170 (Moderate–Poor border) by 8 AM tomorrow, mainly because early-morning traffic is lighter than evening.

For your asthma profile, that's a modest improvement but still warrants the same precautions — mask on, inhaler accessible. If you can travel before 8 AM rather than during the 9–10 AM peak, conditions are typically a little better.

Would you like me to send you a quick check the morning of your return trip with the live numbers?

7.3 Sample RAG Pipeline — What Gets Retrieved and How

Using the conversation above as the working example, the pipeline executes as follows:

Step A — Retrieved Live Data

RETRIEVED CONTEXT BLOCK (LIVE AQI)

```
{
  "route_zones": [
    {"zone": "Chembur", "station": "SAFAR-Chembur", "aqi": 198,
     "category": "Poor", "primary_pollutant": "PM2.5"},
    {"zone": "Sion-Kurla", "station": "SAFAR-Sion", "aqi": 205,
     "category": "Very Poor", "primary_pollutant": "PM2.5"},
    {"zone": "BKC", "station": "SAFAR-BKC (nearest: Bandra)", "aqi": 140,
     "category": "Moderate", "primary_pollutant": "PM10"}
  ],
  "worst_zone": "Sion-Kurla",
  "worst_aqi": 205,
  "active_alerts": [],
  "forecast_trend": "Improving slightly overnight; ~160-170 by 8AM tomorrow",
  "data_timestamp": "2026-06-24T17:40:00+05:30"
}
```

Step B — Retrieved Knowledge Base Snippets

- WHO guideline excerpt for AQI 150–200 (Poor / Unhealthy for Sensitive Groups): respiratory-sensitive individuals should limit prolonged outdoor exertion and consider mask use.
- Mumbai-specific note: Eastern Express Highway and Sion-Kurla stretch typically show elevated readings during the 5–8 PM traffic peak due to congestion-linked vehicular emissions.
- Asthma-specific protocol: at AQI 150–220, recommend N95 mask, inhaler accessibility, and avoidance of unshielded outdoor waiting areas.

Step C — Final Generated Output

Granite receives the live data block, the knowledge snippets, and the user's stored profile (asthma: mild; age group: adult) inside Prompt Template 1, and produces the structured, conversational advisory shown in the chat transcript above — never inventing a number that wasn't in the retrieved context, and explicitly softening or escalating its tone based on the user's specific health condition.

8. Responsible AI Considerations

BREATHE SMART MUMBAI sits at the intersection of public health and AI — a domain where responsible design is not optional polish but a core requirement, as mandated in the internship's Responsible AI guidelines. The four pillars below are addressed with concrete, Mumbai-specific mechanisms, not generic statements.

8.1 Fairness

Mumbai-Specific Challenge: A tool that works best in English and for smartphone owners would systematically underserve the lower-income, Marathi/Hindi-speaking residents of Govandi, Dharavi, and Mankhurd — precisely the populations facing the worst air quality.

- Equal-quality multilingual support: Marathi and Hindi outputs are tested separately by native speakers (Section 5, Stage 5, Phase 1) against the same quality bar as English, not treated as an afterthought translation.
- Low-bandwidth access: A WhatsApp-bot interface (alongside the web app) ensures street vendors, daily wage workers, and elderly users without smartphones-with-apps can still access advisories on basic data connections.
- Coverage gap transparency: For zones with sparse SAFAR coverage (parts of Govandi, Mankhurd, inner Dharavi), the system explicitly flags that it is using the nearest available station rather than silently assuming uniform air quality across all income groups and neighborhoods.
- No paywall: Core advisory features remain free, since the populations most exposed to industrial pollution in Chembur, Mahul, and Govandi are also among the least able to pay for premium air-quality apps.

8.2 Transparency

Mumbai-Specific Challenge: A user deciding whether to mask up for a Govandi-Kurla commute needs to trust where the AQI number came from and why the advice says what it says — not just receive an opaque recommendation.

- Source citation in every advisory: Each response names the specific SAFAR or CPCB station the AQI figure came from (e.g., "According to SAFAR Chembur station...") so users can independently verify on the SAFAR app if they wish.

- Explicit data freshness: If a station's reading is more than 3 hours old, the system flags it as potentially stale rather than presenting it as current (see Prompt Template 2 in Section 5).
- Visible reasoning structure: Every advisory follows the same disclosed structure — AQI summary, personalized risk, recommendation, precautions — so users learn to read and question the system's logic over time rather than treating it as a black box.
- Clear AI disclosure: The assistant identifies itself as an AI tool drawing on public AQI and health-guideline data, not a doctor or a government authority.

8.3 Ethics

Mumbai-Specific Challenge: Wrong or overconfident medical-adjacent advice to an elderly Borivali resident or a Chembur asthma patient could cause real harm — either by under-warning during a Severe AQI event or over-warning to the point users stop trusting any alert.

- Hard-coded safety floor: The system prompt forbids fabricating AQI values and mandates explicit "data unavailable" responses rather than guessing — preventing confident-sounding but false reassurance.
- Escalation protocol: For AQI above 300 (Severe/Hazardous, as can occur near Mahul or during Diwali), the system always surfaces the MCGM emergency health helpline (1916) and recommends avoiding outdoor exposure, regardless of user profile.
- No diagnostic overreach: The assistant gives air-quality-linked precautions, not medical diagnoses — it explicitly defers to the user's doctor for any decision beyond commute-timing and protective measures, and says so directly when a query strays into diagnosis territory.
- Alert-fatigue calibration: Caution thresholds are deliberately tiered (Section 5, Stage 5 ethical issues table) so that "avoid travel" language is reserved for genuinely severe conditions, preserving the credibility of warnings for outdoor workers in BKC and Dharavi who cannot afford to ignore them when it matters.

8.4 Privacy

Mumbai-Specific Challenge: Health conditions (asthma, heart disease) and daily commute routes are sensitive personal data — a breach could expose a person's medical status or reveal predictable movement patterns between home in Govandi and work in BKC.

- On-device profile storage: Health profile data (age group, conditions, regular route) is stored locally on the user's device by default, not on a central server.
- Anonymous session mode: Users who don't want to save any profile can enter health context fresh for a single session — nothing is retained after the conversation ends.
- No location tracking: The system only uses the route a user explicitly types or speaks (e.g., "Chembur to BKC") — it does not access GPS location continuously or build a movement history.
- Minimal data retention for feedback: Feedback collected for system improvement (Section 5, Stage 5, Phase 2) is aggregated and de-identified before being used to refine zone-to-advice mapping.

9. Expected Impact Statement

9.1 What Changes at City Scale

Mumbai's 20+ million residents currently share one resource for air quality awareness: a dashboard of numbers with no personal translation. If BREATHE SMART MUMBAI were adopted at scale — for instance, integrated as an MCGM-endorsed WhatsApp service or embedded into existing transit apps — the core shift is moving air quality from an abstract civic statistic to a daily, personal, actionable decision point for millions of commutes.

Impact Dimension	Expected Effect at Scale
Daily commute decisions informed	Even modest adoption among the 7.5 million daily suburban rail riders and millions of BEST bus/two-wheeler users could mean hundreds of thousands of commutes per day are planned with real AQI awareness rather than none.
Vulnerable population protection	Mumbai's an estimated 1.2 million diagnosed asthma patients and a large elderly population (60+) gain a free, personalized early-warning layer that currently does not exist for them at all.
Mask and precaution uptake	Personalized framing ("for your asthma, today's AQI 205 near Kurla is in the danger zone") is designed to shift behavior more effectively than generic "AQI is high" alerts, which testing in Section 5 aims to validate directly.
Reduced acute health events	Earlier, condition-specific warnings during Severe AQI spikes (Diwali, Mahul/Chembur industrial events) could reduce avoidable asthma attacks and cardiac stress episodes tied to acute exposure — a measurable target for post-pilot evaluation.
Outdoor worker harm reduction	Over 500,000 construction workers and countless street vendors who cannot simply avoid pollution gain practical harm-reduction guidance (mask type, hydration, shade breaks) tailored to their reality, rather than advice that assumes they can stay indoors.

9.2 Who Benefits and How

- Asthma & heart condition patients: Receive condition-specific go/caution/avoid guidance tied to their exact commute, rather than a generic city-wide AQI number with no personal relevance.
- Elderly citizens (60+): Get simplified, gentler-toned advisories in their preferred language (Marathi/Hindi), closing an accessibility gap left by English-only dashboards like SAFAR's current interface.
- Outdoor and daily-wage workers: Gain realistic harm-reduction advice instead of impractical "stay indoors" messaging that ignores their economic constraints.
- Parents and school children: Can check whether a Chembur, Govandi, or Kurla school-route commute warrants a mask or a schedule shift before sending children out.
- Public health bodies (MCGM, MPCB): Gain an additional, low-cost behavioral-change channel that complements — rather than duplicates — their existing monitoring infrastructure, and can use aggregated, de-identified feedback to spot underserved zones.

9.3 Social, Environmental, and Public Health Impact

Impact Type	Description
Public Health	Earlier, personalized warnings for high-risk groups can reduce preventable respiratory and cardiovascular events linked to acute pollution exposure — directly advancing SDG Target 3.4 and 3.9

	outcomes for Mumbai's most exposed neighborhoods (Mahul, Chembur, Govandi).
Social Equity	By prioritizing free access, regional-language support, and a WhatsApp interface, the tool closes an information gap that currently favors English-speaking, smartphone-app-savvy, higher-income residents — extending protection to street vendors, daily-wage workers, and lower-income communities in Dharavi and Govandi.
Environmental Awareness	Repeated, personalized exposure to "why" a route is risky (industrial corridor, construction dust, Diwali smoke) builds long-term public awareness of pollution sources, potentially strengthening civic support for emission-reduction policy around Taloja MIDC, Mahul, and Chembur.
Behavioral Shift at Scale	Aggregated, de-identified usage patterns (which routes/times are most queried as risky) could give MCGM and MPCB a previously unavailable, crowd-sourced signal of where commuters perceive — and experience — the worst air quality, complementing fixed-station monitoring.

None of these figures are claimed as guaranteed outcomes — they represent the scale of the population the tool is designed to reach and the categories of benefit the pilot testing in Section 5 is specifically structured to measure and validate before any city-wide rollout.

End of Document | BREATHE SMART MUMBAI
1M1B AI for Sustainability Virtual Internship | IBM SkillsBuild & AICTE